Review 3: Final Manuscript & Experimental Evaluation

Hospital Emergency Resource Manager Using Operating System Concepts
Nitin Sunil Holkar (24BCE2819), Gaurav Charpe (24BCT0326), Aditya Phogaat | Faculty: Dr. V. Ezhil Arasi
School of Computer Science and Engineering, Vellore Institute of Technology, Vellore

Abstract & Summary

This paper presents the Hospital Emergency Resource Manager (HERM), an operating-system-based concurrent framework for coordinating scarce emergency clinical resources. Patients are modeled as processes, and medical infrastructure is synchronized using counting semaphores, mutex locks, hybrid preemptive priority scheduling with aging, Banker's deadlock avoidance, and WFG preemption recovery. Controlled benchmarks on high-contention surge workloads demonstrate a 91.6% reduction in critical patient waiting time.

Experimental Evaluation & Comparative Benchmarks

Metric	FCFS	Non-Preemptive	Preemptive + Aging (HERM)	No Banker's
Critical (L1) Avg Wait	8.67 s	3.33 s	0.67 s	0.67 s
Serious (L2) Avg Wait	3.50 s	4.00 s	3.50 s	3.50 s
Moderate (L3) Avg Wait	3.50 s	3.50 s	4.00 s	4.00 s
Stable (L4) Avg Wait	2.67 s	4.67 s	4.33 s	4.33 s
Overall Avg Wait Time	7.70 s	7.10 s	4.40 s	4.40 s
Starvation Count (W>=15)	0	0	0	0
Deadlocks Detected	0	0	0 (Avoided)	1 (Recovered)

Key Empirical Findings

1. Preemption Slashes Critical Latency: Level 1 Critical patients wait only 0.67s under HERM compared to 8.00s under FCFS and 3.33s under Non-Preemptive Priority.

2. Aging Eliminates Starvation: While non-preemptive priority prolongs lower-urgency delays, dynamic aging caps waiting time by promoting long-waiting Level 3 and 4 patients to Level 2.

3. Provable Deadlock Control: Banker's safety state proactively avoids 100% of deadlocks. When intentionally disabled, the WFG DFS detector catches circular dependencies within 1 clock tick and resolves them via preemption.

GitHub Repository & Verification

The complete implementation in C with POSIX threads is prepared for review submission:

Repository URL: https://github.com/gauravcharpe/hospital-emergency-resource-manager

Compilation: make (compiles herm_cli and herm_server)

Execution: ./herm_cli (Terminal CLI) | ./herm_server (Pure C Web Server) | ./herm_cli --benchmark